Group Activity: Annual Carbon Emissions

Student worksheet RBE201TC

Team _____ Date _____

Use a calculator. Each member calculates one person's annual emissions, then the team compares and checks the results. Complete the blank workings and response areas.

Activity instructions

1. Assign people. Work in groups of 4–5 (4 recommended). Assign Lin, Chen, Zhou and Wang to different members. In a five-person group, two members independently calculate the same case.

2. Calculate independently. Each member calculates every category for one person, recording annual quantities, units and emissions.

3. Compare and cross-check. Exchange workings, check assumptions and units, then rank all four people from highest to lowest annual emissions.

4. Present individually. Each member shows their calculation steps and total, identifies the largest source and proposes one reduction.

Emission factors

Source: carbon_footprint_calculator.xlsx (course teaching calculator). Use these fixed classroom factors for this activity.

Category	Usage unit	Factor in kg CO_2e per unit
Electricity	kWh	0.582
Shared bike	km	0.01
Gasoline car	km	0.17
High-speed rail	passenger-km	0.03
Electric vehicle	km	0.15
Domestic flight	passenger-km	0.15
Bus	passenger-km	0.05
Household waste	kg	0.6
Subway	passenger-km	0.04
Water supply	L	0.001

Calculation guide

Shared annual usage. Annual household usage ÷ number of residents

Daily or weekly activity. Daily quantity × stated days; or weekly quantity × stated weeks

Monthly activity. Monthly quantity × 12

Return journeys. One-way distance × 2 × number of round trips

Distance already given as a round trip. Round-trip distance × number of round trips (do not multiply by 2 again)

Household waste. Daily waste (kg) × 365

Annual category emissions. Annual quantity × compatible emission factor = kg CO_2e/year

Total annual emissions. Add the emissions of every category

Largest-source share (%). Largest category emissions ÷ total annual emissions × 100

Reduction (%). (Original total − revised total) ÷ original total × 100

Units and shared assumptions

CO_2e means carbon dioxide equivalent: greenhouse gases expressed on a common warming-impact basis. kg is a unit of mass. kWh measures electricity use; L measures water volume.

1 passenger-km means one person travelling 1 km. A factor of 0.05 kg CO_2e/passenger-km assigns 0.05 kg CO_2e to each passenger-km travelled.

Use 365 days/year and the stated weeks or travel days. Divide shared household electricity and water by the number of residents. Unlisted transport = 0.

Takeaway packaging is already included in household waste. Do not calculate it separately or add food emissions. Keep extra digits while calculating; round final emissions to 2 decimals.

For each category, show the annualisation and household-sharing steps, then multiply by the factor. Record a true zero for unused transport.

Group comparison

Compare all four people after checking one another's calculations. Enter rank 1 for the highest annual total.

Person	Calculated by	Annual total kg CO_2e/year	Rank 1 = highest	Largest source	Checked by
Lin					
Chen					
Zhou					
Wang					

Lin, 16, High School Student

Calculated by _____ Checked by _____

Lives with two parents. The household uses 3,600 kWh of electricity and 108,000 L of water annually, shared equally among three people.

Takes the bus to school five days a week for 40 weeks each year. The one-way journey is 6 km.

Uses a shared bike on 40 weekends each year, travelling 4 km per round trip.

Makes two round trips by high-speed rail each year to visit relatives. Each one-way journey is 500 km.

Produces 0.4 kg of household waste daily and orders one takeaway meal per week for 50 weeks each year.

Show the annual quantity and its calculation, then the factor multiplication and emissions result. Use the factors and assumptions on the preceding pages.

Category and usage unit	Annual quantity and workings	Factor kg CO_2e per unit	Emissions workings and result kg CO_2e/year
Electricity (kWh)			
Water supply (L)			
Gasoline car (km)			
Electric vehicle (km)			
Bus (passenger-km)			
Subway (passenger-km)			
Shared bike (km)			
High-speed rail (passenger-km)			
Domestic flight (passenger-km)			
Household waste (kg)			

Total annual emissions (kg CO_2e/year) _____

Largest source _____ Share of total (%) _____

Chen, 20, University Student

Calculated by _____ Checked by _____

Shares a dormitory room with three other students. The room uses 2,400 kWh of electricity and 120,000 L of water annually, shared equally among four people.

Uses a shared bike on 200 days each year, travelling 4 km per day in total.

For 40 weeks each year, takes one bus round trip of 10 km and two subway round trips of 20 km each per week.

Makes two round trips by high-speed rail each year to visit home. Each one-way journey is 800 km.

Produces 0.5 kg of household waste daily and orders four takeaway meals per week for 50 weeks each year.

Show the annual quantity and its calculation, then the factor multiplication and emissions result. Use the factors and assumptions on the preceding pages.

Category and usage unit	Annual quantity and workings	Factor kg CO_2e per unit	Emissions workings and result kg CO_2e/year
Electricity (kWh)			
Water supply (L)			
Gasoline car (km)			
Electric vehicle (km)			
Bus (passenger-km)			
Subway (passenger-km)			
Shared bike (km)			
High-speed rail (passenger-km)			
Domestic flight (passenger-km)			
Household waste (kg)			

Total annual emissions (kg CO_2e/year) _____

Largest source _____ Share of total (%) _____

Zhou, 35, Engineer

Calculated by _____ Checked by _____

Lives alone and uses 2,400 kWh of electricity and 45,000 L of water annually at home.

Drives a gasoline car alone to work on 220 days each year, travelling 40 km per day in total.

Makes four domestic round trips by air each year for business. Each one-way flight is 1,000 km.

Makes one round trip by high-speed rail each year for a holiday. Each one-way journey is 500 km.

Produces 0.8 kg of household waste daily and orders five takeaway meals per week for 50 weeks each year.

Show the annual quantity and its calculation, then the factor multiplication and emissions result. Use the factors and assumptions on the preceding pages.

Category and usage unit	Annual quantity and workings	Factor kg CO_2e per unit	Emissions workings and result kg CO_2e/year
Electricity (kWh)			
Water supply (L)			
Gasoline car (km)			
Electric vehicle (km)			
Bus (passenger-km)			
Subway (passenger-km)			
Shared bike (km)			
High-speed rail (passenger-km)			
Domestic flight (passenger-km)			
Household waste (kg)			

Total annual emissions (kg CO_2e/year) _____

Largest source _____ Share of total (%) _____

Wang, 68, Retired

Calculated by _____ Checked by _____

Lives with a partner. The household uses 2,400 kWh of electricity and 60,000 L of water annually, shared equally between two people.

Usually walks for local journeys. Takes three bus round trips per week, each covering 8 km, for 50 weeks each year.

Makes one round trip by high-speed rail each year to visit relatives. Each one-way journey is 600 km.

Produces 0.35 kg of household waste daily and orders two takeaway meals per month.

Show the annual quantity and its calculation, then the factor multiplication and emissions result. Use the factors and assumptions on the preceding pages.

Category and usage unit	Annual quantity and workings	Factor kg CO_2e per unit	Emissions workings and result kg CO_2e/year
Electricity (kWh)			
Water supply (L)			
Gasoline car (km)			
Electric vehicle (km)			
Bus (passenger-km)			
Subway (passenger-km)			
Shared bike (km)			
High-speed rail (passenger-km)			
Domestic flight (passenger-km)			
Household waste (kg)			

Total annual emissions (kg CO_2e/year) _____

Largest source _____ Share of total (%) _____